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Agile & DevOps Lab Assignment 3

Install Maven:

```
PS C:\Users\Maheshwar> mvn --version
Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)
Maven home: C:\ProgramData\chocolatey\lib\maven\apache-maven-3.9.11
Java version: 17.0.16, vendor: Amazon.com Inc., runtime: C:\Program Files\Amazon Corretto\jdk17.0.
Default locale: en_IN, platform encoding: Cp1252
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

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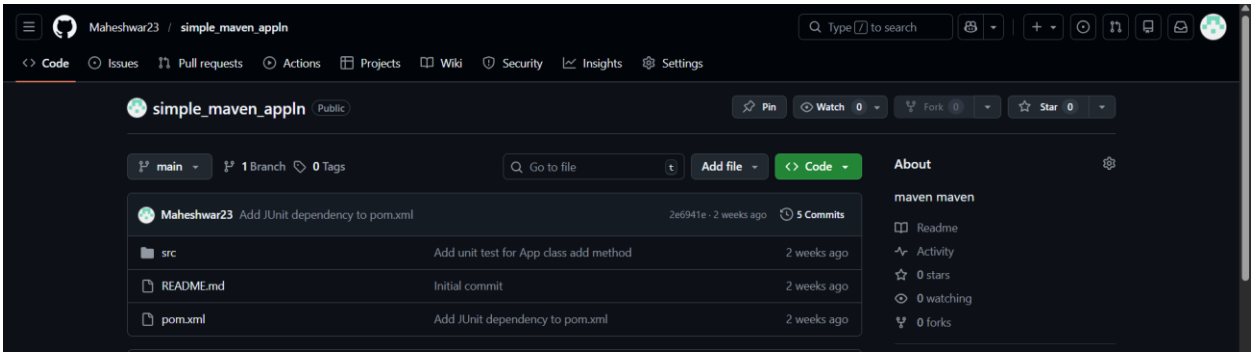
GITHUB CREATE A NEW REPOSITORY:SIMPLE-MAVEN-APPLICATION:



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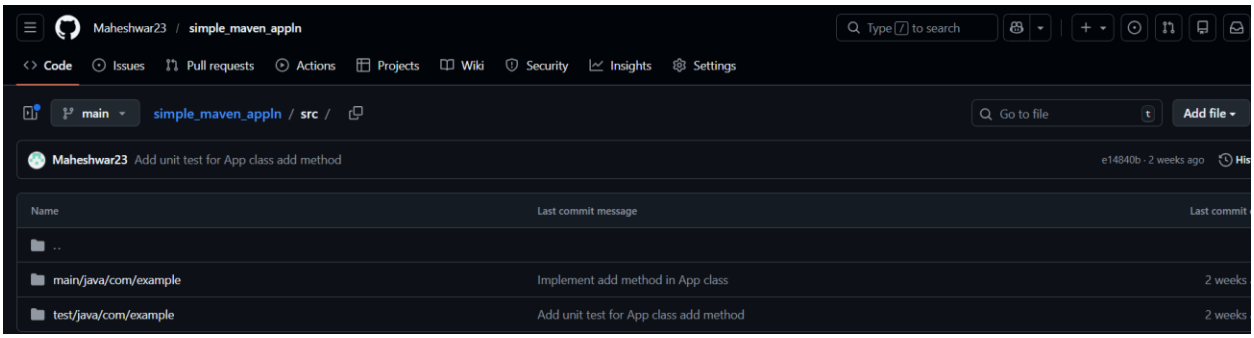

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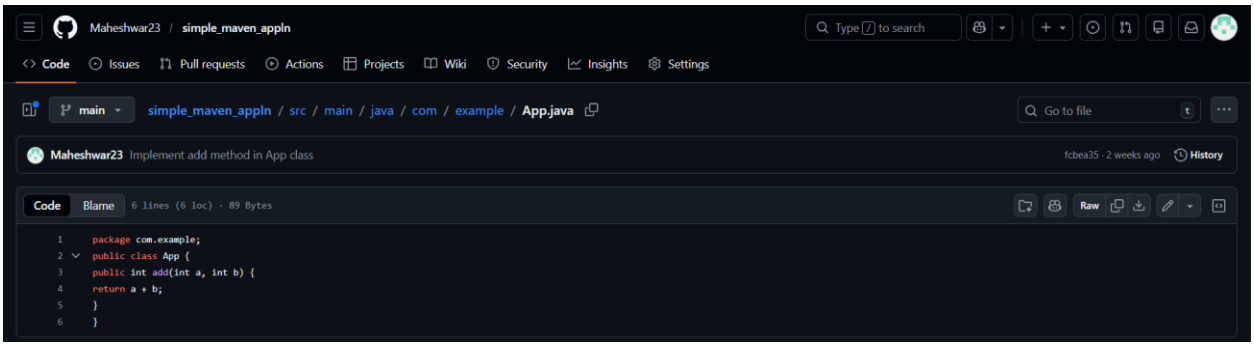
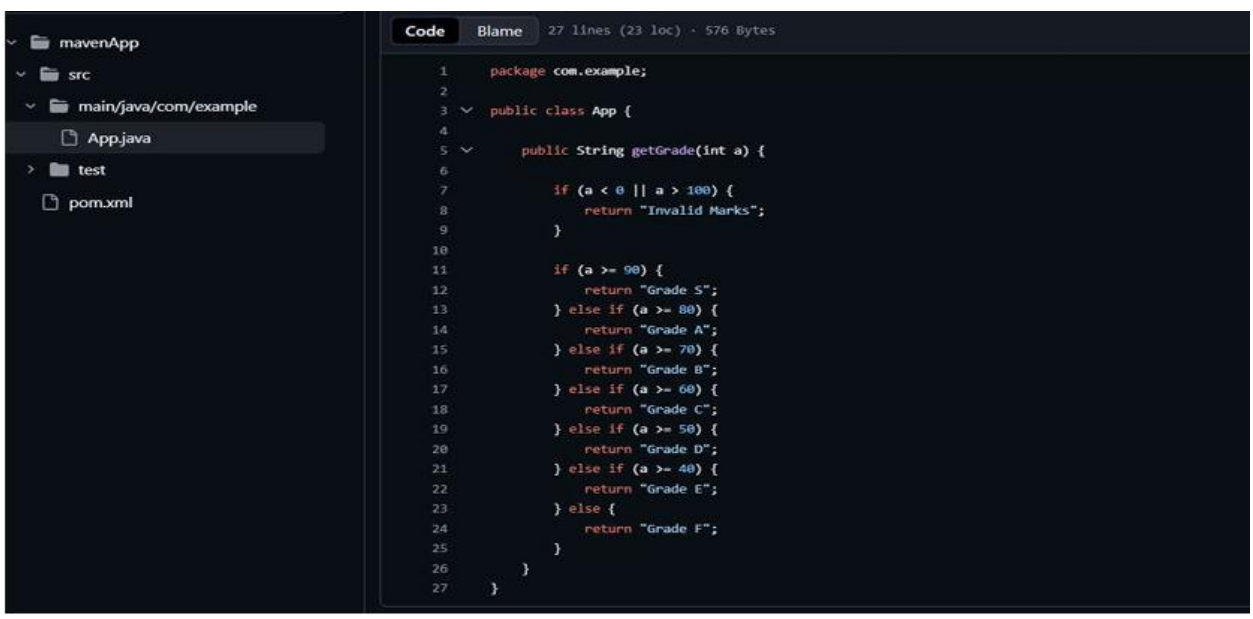


For student grade calculations:App.java:



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AppTest.java:

SIMPLE-MAVEN-APPLICATION / mavenApp / src / test / java / com / example / AppTest.java

Code Blame 40 lines (32 loc) · 1.09 KB

```java
import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    @Test
    public void testGrades() {
        App app = new App();

        assertEquals("Grade S", app.getGrade(95));
        assertEquals("Grade A", app.getGrade(85));
        assertEquals("Grade B", app.getGrade(72));
        assertEquals("Grade C", app.getGrade(60));
        assertEquals("Grade D", app.getGrade(55));
        assertEquals("Grade E", app.getGrade(45));
        assertEquals("Grade F", app.getGrade(30));
    }

    @Test
    public void testEdgeCases() {
        App app = new App();

        assertEquals("Grade S", app.getGrade(100));
        assertEquals("Grade F", app.getGrade(0));
    }

    @Test
    public void testBoundaryValues() {
        App app = new App();

        assertEquals("Grade E", app.getGrade(40));
        assertEquals("Grade D", app.getGrade(50));
        assertEquals("Grade C", app.getGrade(60));
        assertEquals("Grade B", app.getGrade(70));
        assertEquals("Grade A", app.getGrade(80));
        assertEquals("Grade S", app.getGrade(90));
    }
```

Pom.xml:

Maheshwar23 / simple_maven_appln

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

main simple_maven_appln / pom.xml

Maheshwar23 Add JUnit dependency to pom.xml 2e6941e · 2 weeks ago History

Code Blame 18 lines (17 loc) · 523 Bytes

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>simple-maven-app</artifactId>
<version>1.0-SNAPSHOT</version>
<dependencies>
 <dependency>
 <groupId>junit</groupId>
 <artifactId>junit</artifactId>
 <version>4.13.2</version>
 <scope>test</scope>
 </dependency>
</dependencies>
</project>
```

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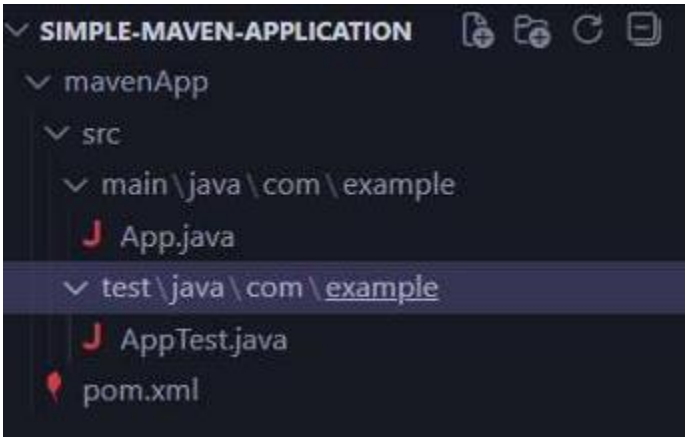
`

Git Clone in VS code:



Output of mvn clean test:

```
it4-3.2.5.jar (26 kB at 404 kB/s)
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.039 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  9.050 s
[INFO] Finished at: 2026-03-02T22:23:15+05:30
[INFO] ------------------------------------------------------------------------
```

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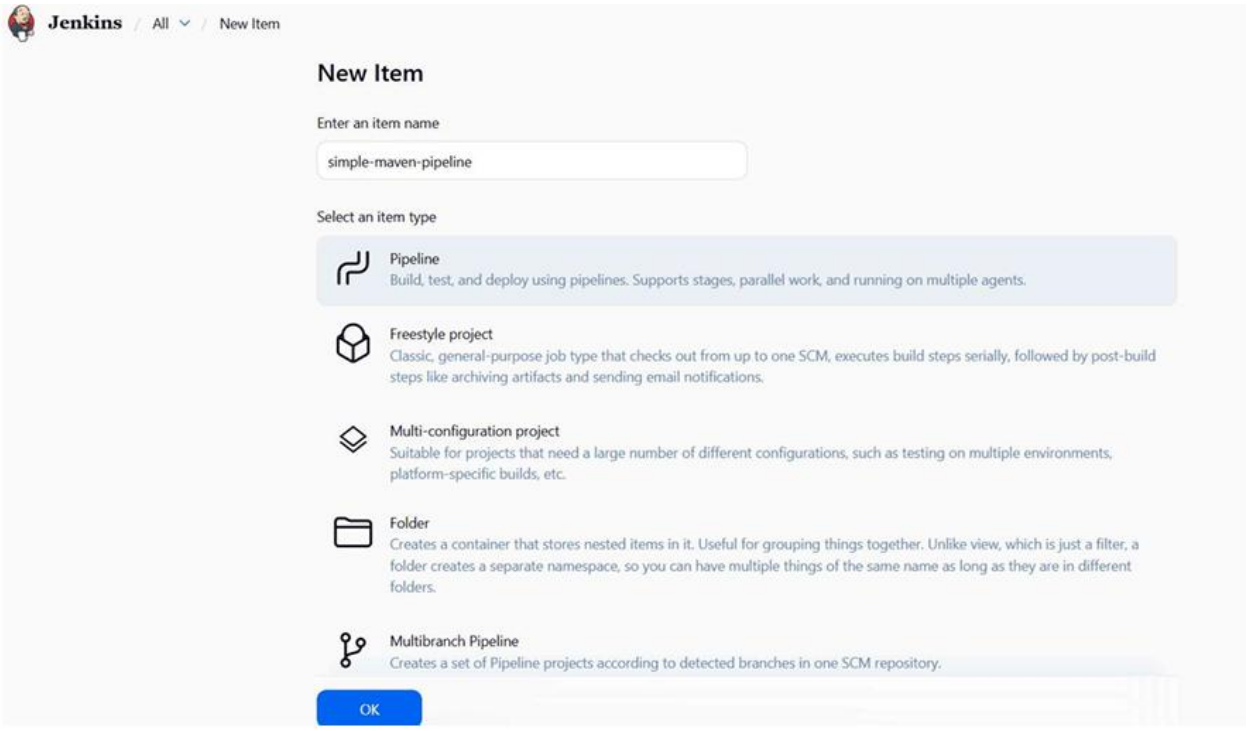
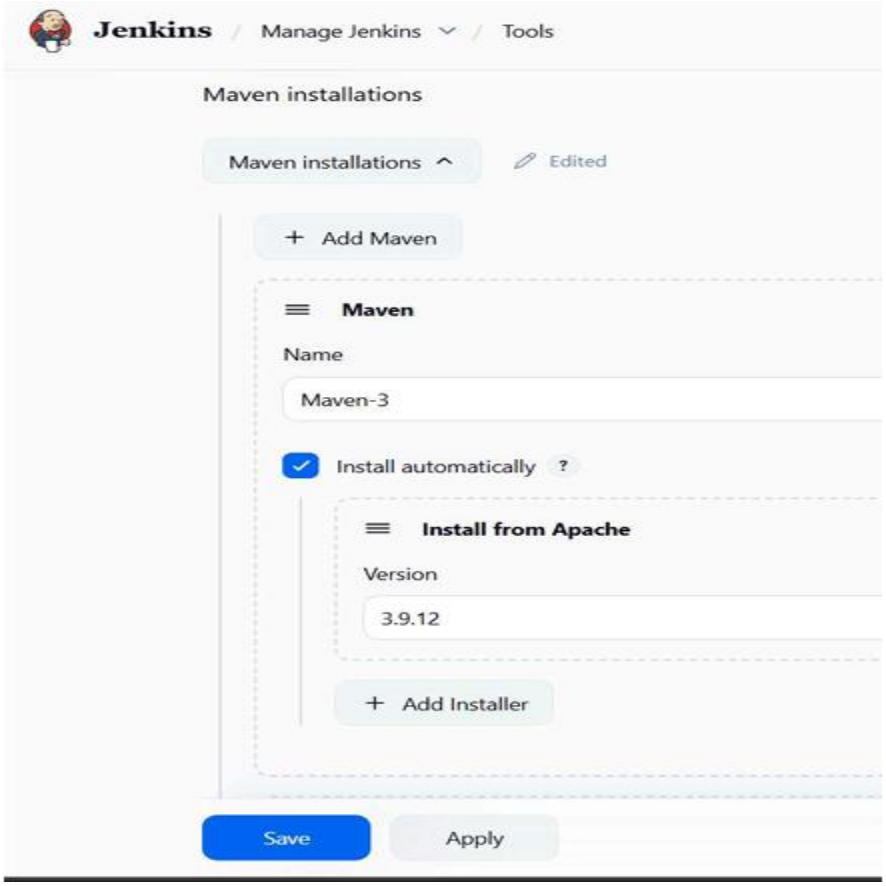
Move code in Jenkins:

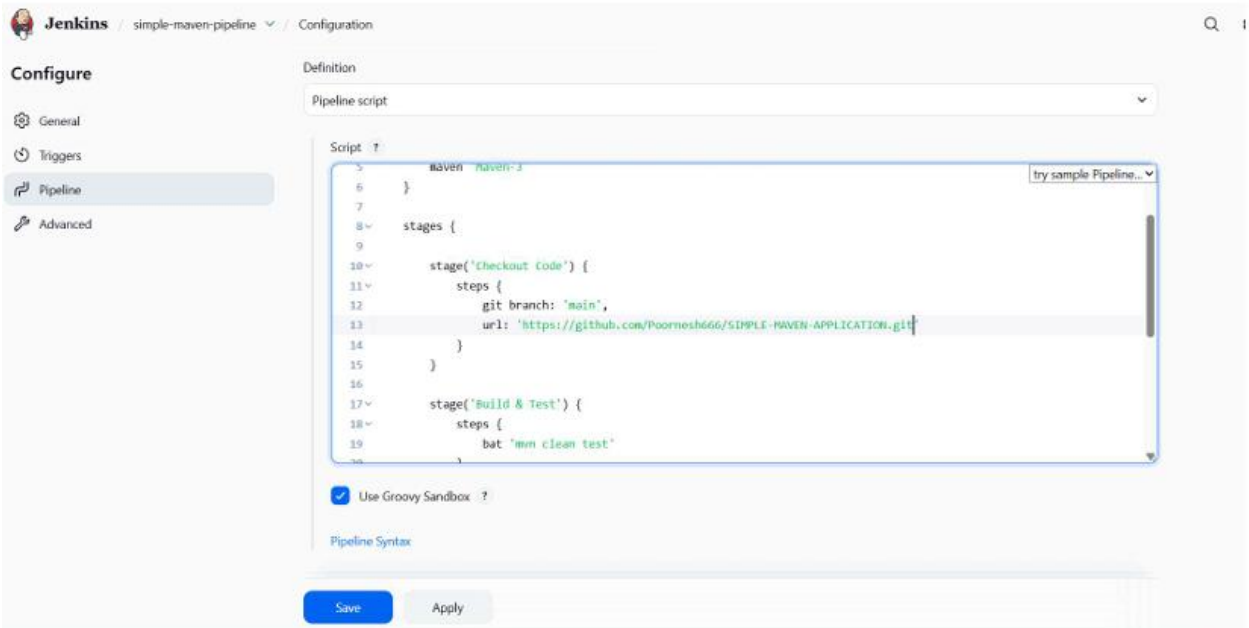


Maven Configurations in Jenkins:

`



Create Pipeline Script:



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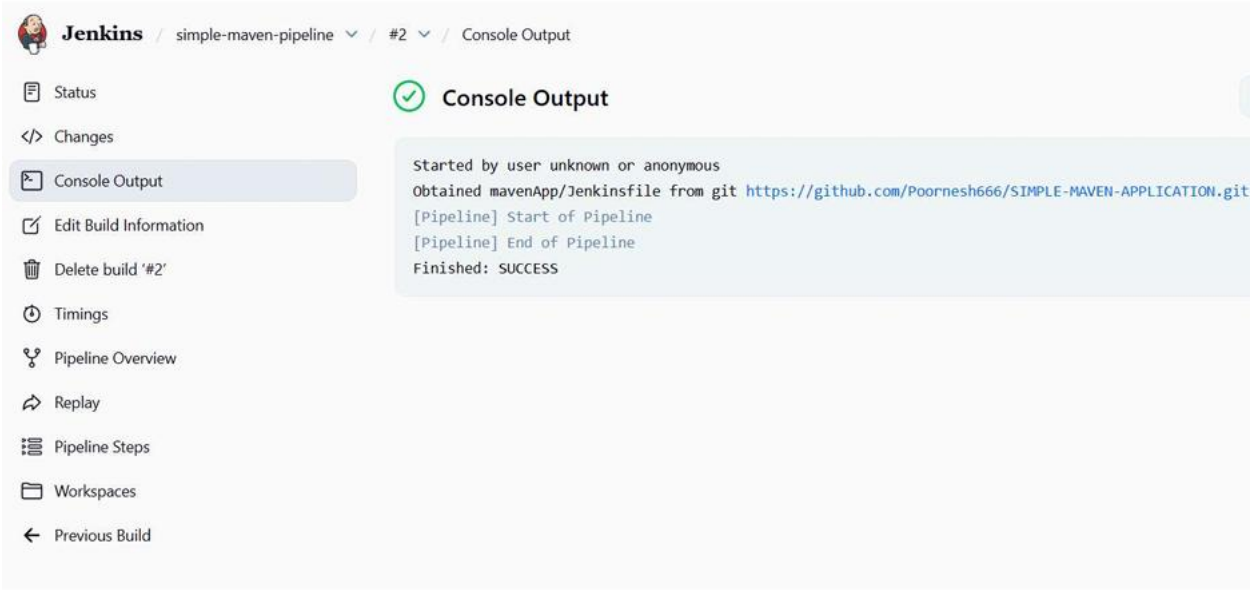
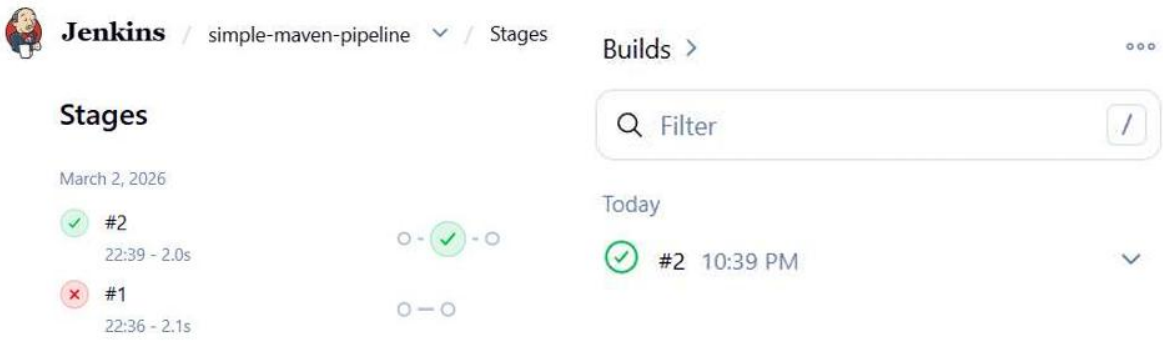
Successful Build Completion:





`